

Department of CSE, IIT Guwahati

CS 341 Computer Networks Quiz 2 Max.Marks: 20 Duration: 50 min

- **The question paper is divided into two sections.**
- **Please write all answers for Section A in the designated space of the question booklet and all answers for Section B in the answer booklet provided.**
- **Answer all questions precisely. Answers longer than required will not be considered.**
- **For all numerical questions, show your calculation steps clearly.**

SECTION A

1. **[1]** Name the two most important attributes in a BGP advertisement along with the prefix. 0/1 marks

AS-PATH
NEXT-HOP

2. **[1]** Mention the two types of ICMP messages received at the sending host executing the Traceroute program. 0/1 marks

The following ICMP messages are used; TTL Expired, Dest Port Unreachable

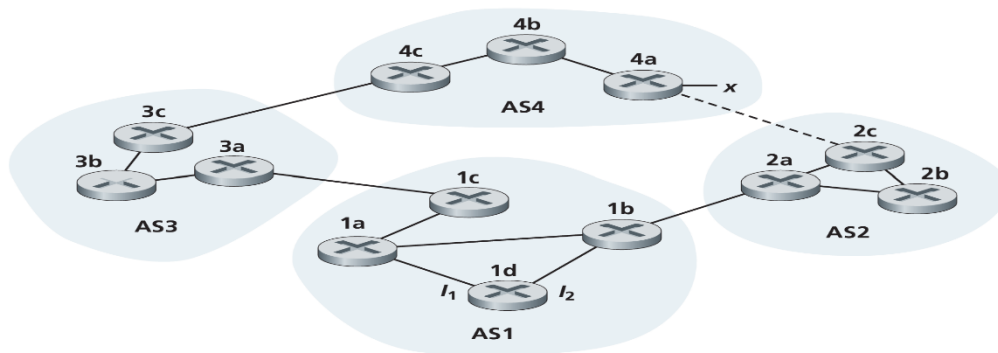
3. **[1]** In Ethernet, if the NIC detects collision, the transmission is aborted and a special signal called _____ is sent. 0/1 marks

Sol : Jamming signal or Jam signal

4. **[1]** What is the main requirement for nodes implementing slotted ALOHA, that is not necessary in pure ALOHA or CSMA? 0/1 marks

Time synchronization between the transmitting stations

5. **[3]** In the network shown below, suppose AS3 and AS2 are running OSPF for their intra-domain routing, while AS1 and AS4 are running RIP for their intra-domain routing. Suppose eBGP and iBGP are used for the inter-domain routing. Initially suppose that the link between AS2 and AS4 (link in dotted line) is not active.



a. Router 3c learns about prefix x from which routing protocol?

eBGP

b. Router 3a learns about x from which routing protocol?

iBGP

A and B together 0/1 marks

c. Router 4c learns about x from which routing protocol? 0/1 marks

RIP

d. Assume that the link between AS4 and AS2 becomes active and router 1d learns about x from both AS3 and AS2, which interface (among I₁ and I₂) is selected for routing to x and why? 0/1 marks

**I₂,
Hot potato routing,** Routers always take the exit BGP router via the least cost intra-AS route.

6. **[4]** Mention if the following statements are TRUE or FALSE with supporting justification,

not exceeding 1-2 sentences. No credit without justification.

- a. In OSPF routing, a router's link state information is sent only to directly connected neighbouring nodes. 0/1 marks

False. In OSPF, link-state information is **flooded** to all routers within an area, not just directly attached neighbors, ensuring a consistent network view across the area.

- b. When a BGP router receives an advertised path over an eBGP connection, it always forwards the advertisement to the other BGP routers after adding its own AS id. 0/1 marks

False. When a BGP router receives advertisement, it applies **import policy filter** that could prevent re-advertising the path.

- c. In the CSMA protocol, if all nodes perform carrier sensing before attempting to transmit, collisions can be completely avoided. 0/1 marks

False, collisions cannot be completely avoided in the Carrier Sense Multiple Access (CSMA) protocol, but it can significantly reduce their occurrence.

- d. In random access MAC protocol, the exponential back-off mechanism reduces the probability of collisions. 0/1 marks

True, in a random access MAC protocol, the exponential back-off mechanism effectively reduces the probability of collisions by introducing a random delay when a node detects a collision, which helps to spread out retransmission attempts and minimize the chance of multiple nodes trying to transmit at the same time

(P.T.O)

SECTION-B

7. [3] Consider an Ethernet segment with a transmission speed of 10^7 bits/sec and a maximum segment length of 500 m. If the speed of propagation of the signal in the medium is 2×10^8 meters/sec, then calculate the minimum frame size (in bits) required for collision detection.

Sol: For collision detection, the transmission time for the frame must be at least as long as the round-trip propagation delay. The minimum frame size L (in bits) can be calculated by

Correct equation only : 1 marks

Required transmission time $T_x \geq \text{round-trip propagation delay}$

$$L/B \geq 2 * d/v$$

$$L = 2 * d/v * B = 50 \text{ bits}$$

Thus, the minimum frame size required for collision detection is 50 bits.

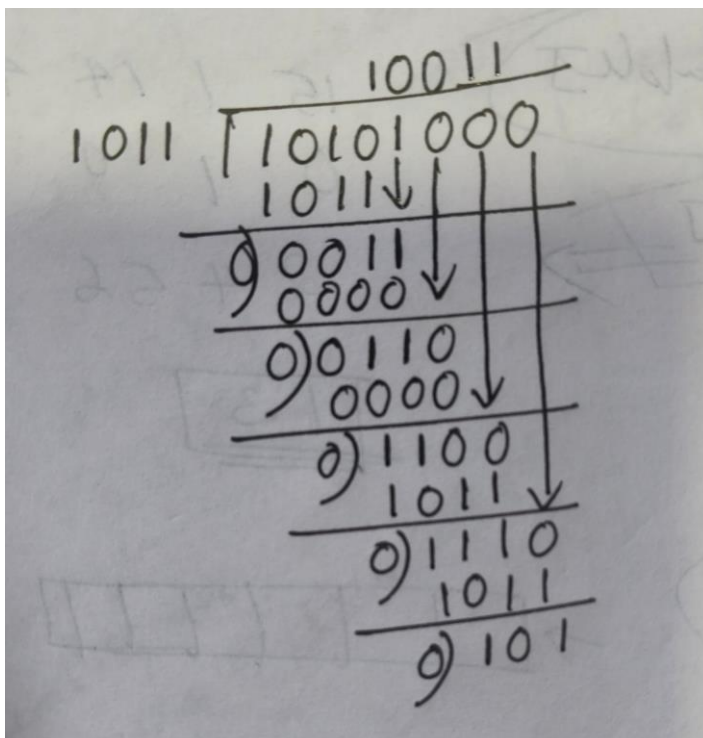
8. [3] Consider the CRC-based error detection scheme with the generator polynomial $x^3 + x + 1$. Suppose the message 10101 is to be transmitted. What is the bit sequence transmitted?

Identification of generator polynomial only: 1 mark

Sol:

Generator polynomial: $x^3 + x + 1$

This corresponds to the binary divisor $G(x) = 1011$



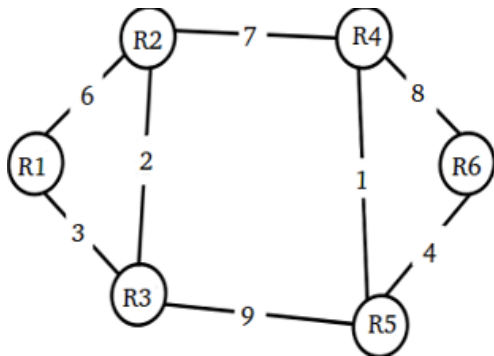
Transmitted bit : 10101101

9. [3] Consider a network of six routers connected with weighted bi-directional links shown below.

1 mark: Provided **1 correct link only**.

3 marks: Identified **both correct links** with a complete table for DV/MST.

0 marks: Does not meet the above criteria.



The routers use distance vector routing algorithm to update their routing tables. After the routing tables at all the nodes stabilize, which links in the network will never be used for carrying any packets? Use the notation $R1-R2$ to indicate link between $R1$ and $R2$.

Shortest Distances from R_1 to R_2, R_3, R_4, R_5 and R_6

$R_1(5, 3, 12, 12, 16)$

Links used: $R_1 - R_3, R_3 - R_2, R_2 - R_4, R_3 - R_5, R_5 - R_6$

Shortest Distances from R_2 to R_3, R_4, R_5 and R_6

$R_2(2, 7, 8, 12)$

Links used: $R_2 - R_3, R_2 - R_4, R_4 - R_5, R_5 - R_6$

Shortest Distances from R_3 to R_4, R_5 and R_6

$R_3(9, 9, 13)$

Links used: $R_3 - R_2, R_2 - R_4, R_3 - R_5, R_5 - R_6$

Shortest Distances from R_4 to R_5 and R_6

$R_4(1, 5)$

Links used: $R_4 - R_5, R_5 - R_6$

Shortest Distance from R_5 to R_6

$R_5(4)$

Links Used: $R_5 - R_6$

If we mark, all the used links one by one, we can see that following links are never used.

$R_1 - R_2$

$R_4 - R_6$

Sol : Final DV Table :
